Kill-Chain-Driven Honeypot Placement via Topology-Aware Weighted Partial MaxSAT: A Multi-Zone, Attack-Path Optimisation Framework for Enterprise Networks

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***Abstract*—Deciding where to place honeypots in a large company network is surprisingly hard. The network is split into different security zones, contains thousands of different types of servers, and faces attackers who follow well-known step-by-step attack patterns called kill chains. On top of that, the organisation has a limited budget, and some honeypot types cannot be deployed at the same time because they would conflict with each other.**

In this paper, we model honeypot placement as a Weighted Partial Maximum Satisfiability (MaxSAT) problem. We encode six goals at once: how well we detect threats, how many attack techniques we cover from the MITRE ATT&CK framework, how many attack tactic families we cover, how well we stop attackers early, how well we catch attackers at the end of their path, and how good our coverage is across all attack paths. All hard rules—budget limits, zone isolation, and conflict constraints—are encoded as clauses that must never be violated.

We test our approach on networks ranging from 100 nodes (tiny) to over 5.5 million nodes (very large), simulating five different real-world attack scenarios. Our MaxSAT solver achieves a quality score of $Q = 67.4$ compared to $Q = 52.8$ for the best competing approach ($+27.8\%$), while also providing a mathematical guarantee that its answer is the best possible solution.

***Index Terms*—Honeypot placement, Weighted Partial MaxSAT, kill chain, MITRE ATT&CK, network deception, multi-zone topology, proactive cyber defence, RC2 solver, enterprise network security.**

I. Introduction

A HONEYPOT is a fake computer system designed to look attractive to attackers. When an attacker interacts with it, the defender learns about their methods and can raise an alert. Honeypots are widely used in network security, but their usefulness depends heavily on *where* they are placed. If a honeypot is put in the wrong place, it will never be visited by attackers and the investment is wasted.

Real company networks make this placement decision much harder than it sounds. Consider a typical large organisation. It has an internet-facing area (called a DMZ) where its public web servers sit. It has an internal office network. It might have cloud services. It may also have industrial control systems (called OT/SCADA) that run physical equipment like power plants or factories. Each of these areas has different security rules, different types of computers, and different risks. Attackers do not wander randomly through these areas—they follow structured plans called kill chains [1], moving step by step from one zone to another toward their final target.

The problem is further complicated by three facts. First, the organisation has a limited budget and cannot afford to deploy a honeypot on every machine. Second, certain combinations of honeypot types are incompatible—for example, an industrial control system honeypot deployed in the cloud zone would be useless because the OT zone is physically isolated from the cloud. Third, there are many possible attack paths through the network and an ideal placement should intercept as many as possible.

Existing methods do not handle all these challenges at the same time. Simple rule-based approaches just follow fixed guidelines. Greedy algorithms pick the best-looking option at each step but cannot reverse a bad decision. Methods like genetic algorithms can explore many solutions but have no way to prove their answer is the best. In this paper, we use a formal solver that considers *all* constraints and objectives together and provides a *proven optimal* answer.

Our main contributions are:

1) **A six-goal placement objective** encoded as a single MaxSAT optimisation problem comprising: (1) detection efficiency, (2) coverage of MITRE ATT&CK attack techniques, (3) coverage of MITRE ATT&CK tactic families, (4) catching attackers early in their path, (5) catching attackers at the end of their path for forensic evidence, and (6) covering as many attack paths as possible. Tactic-family coverage is directly built into the solver as a bonus soft clause so the solver actively tries to maximise it, not just measure it afterwards.
2) **A priority system with four levels** for the solver goals. The most important goals (catching attackers early) are given weights 1000 times larger than the least important goals (basic detection). We also encode the key insight that one honeypot can cover multiple attack paths at once—something no greedy method can discover.
3) **A thorough experimental evaluation** across six different network sizes, using a Monte Carlo simulation to model attacker behaviour and testing what happens when threat intelligence updates cause us to re-run the solver with new priorities.

The rest of this paper is organised as follows. Section II discusses related work. Section III defines the problem formally. Section IV presents our MaxSAT model. Section V describes our experiments. Section VI presents the results. Section VII explains why our approach beats the baselines. Section VIII covers limitations and future work. Section IX concludes.

II. Related Work

This section reviews the four bodies of research most relevant to our work: honeypot placement, MITRE ATT&CK-based defence, formal methods for network security, and moving-target defence. For each area we explain what has been done and identify the specific gap that our approach fills. We then present a systematic novelty analysis—a capability matrix (Figure 1) and a supporting argument—that proves no single prior method satisfies all five capability dimensions simultaneously.

A. Honeypot Placement Research

Honeypots have been studied for decades. Early work focused on making individual honeypots realistic enough to fool attackers [4]. Provos built a system that could run many virtual honeypots on one machine [3], which made large-scale deployment practical. Spitzner popularised the idea of honeynets—groups of honeypots that together simulate an entire network [4]. However, most of this early work assumed a human expert would decide where to put the honeypots; there was no automated decision-making process.

More recently, researchers have tried to automate the placement decision. Yuill *et al.* were among the first to frame honeypot placement as a strategic interaction between attacker and defender [7]. Tan *et al.* and Pawlick *et al.* built on this idea using formal game theory to model how rational attackers respond to deception [8], [9]. These approaches are mathematically elegant, but they assume a simple two-player game on a flat, homogeneous network. In reality, enterprise networks have multiple zones with different trust levels, physical air gaps between certain zones, and attackers who can follow many different paths simultaneously. None of the game-theoretic approaches we reviewed handle these structural properties.

Gap identified: Game-theoretic methods cannot encode multi-zone topology (D1), kill-chain step ordering (D2), or honeypot conflict constraints (D4). The solution concept they produce—a Nash equilibrium—is also fundamentally different from a proven weighted optimum.

B. Using MITRE ATT&CK for Defence Planning

MITRE ATT&CK [2] is a publicly available catalogue of the tactics and techniques that real attackers use in practice. It has become the standard reference for both offensive and defensive security research.

Applebaum *et al.* were among the first to use ATT&CK for automated security testing [10]. They built a framework that acts like an automated red team, moving through a network and testing its defences using techniques drawn from the ATT&CK catalogue. Their work showed that ATT&CK can drive realistic attack simulation. However, their goal was to test *existing* defences, not to decide where to place *new* ones.

Hemberg *et al.* used evolutionary (genetic) algorithms to select the best defensive controls from the ATT&CK catalogue [11]. Their approach found good control sets by evolving solutions over many generations and achieves partial ATT&CK coverage. However, they did not include budget constraints, zone topology, or conflict constraints between controls, and genetic algorithms cannot prove that their answer is the best possible one. They also evaluate fitness one objective at a time, which means they cannot discover the cross-path force-multiplier effect that makes our solver 2.1× more efficient on technique coverage.

Gap identified: Both works use ATT&CK to evaluate or select defences, but neither embeds ATT&CK technique weights, stealth scores, or tactic-family bonuses *directly into a solver as live optimisation objectives* (D3). Our work is the first to do this.

C. Using Formal Methods for Network Security

Formal methods use mathematical logic to prove properties of systems. In network security, they have been used to check firewall rules for consistency [12], automatically generate network configurations [13], and verify security policies [14].

Pinisetty *et al.* applied MaxSAT to the problem of selecting the best rules for an intrusion detection system [15]. Their work showed that MaxSAT is a practical tool for security resource allocation. However, they did not model kill-chain structure, zone topology, or honeypot-specific placement objectives.

Integer Linear Programming (ILP) has also been used for security resource allocation [16], [17]. Saha and Gera used ILP to find the optimal locations for security sensors while respecting budget and zone constraints [16], achieving multi-zone support (D1) and proven optimality (D5). However, their model does not encode kill-chain step ordering (D2), does not integrate ATT&CK objectives (D3), and faces a serious scalability problem: ILP becomes very slow as the problem grows and cannot handle the 5.5-million-node networks we evaluate within practical time limits.

Jajodia *et al.* introduced topological analysis of network attack vulnerability [18]. They showed how an attacker's possible paths through a network can be modelled as a directed graph and how chokepoints in that graph are the most valuable places to defend. Our work operationalises this insight formally: chokepoints appear in our model as high-weight path edges, causing the solver to concentrate honeypots there naturally.

Gap identified: No prior work has applied Weighted Partial MaxSAT to honeypot placement with kill-chain-aware objectives and physical zone isolation constraints. The closest method—ILP sensor placement [16]— achieves optimality but misses kill chains (D2), ATT&CK integration (D3), and does not scale beyond tens of thousands of nodes within a 60-second budget.

D. Moving-Target Defence

Moving-target defence (MTD) [19] is a strategy where the defender constantly changes the network configuration to confuse attackers. Instead of keeping the same setup all the time, the defender rotates IP addresses, shuffles network routes, or changes software versions to prevent attackers from building an accurate picture of the network.

MTD and honeypot placement are complementary ideas rather than competing ones: MTD changes the *real* system to create uncertainty, while honeypots add *fake* systems to create traps. Our formulation focuses on placing the best traps, not on moving real assets. However, our adaptive redeployment algorithm (Algorithm 1) adds a dynamic element that bridges toward MTD philosophy: when new threat intelligence arrives and changes the probability of certain attack paths, our solver re-optimises the honeypot placement using the previous solution as a warm start.

Gap identified: MTD work does not address the honeypot placement subproblem, kill-chain ordering (D2), ATT&CK integration (D3), conflict constraints (D4), or optimality guarantees (D5).

E. Novelty Analysis: Related Work vs. This Paper

We now demonstrate systematically that our work is novel by mapping each body of related work to five capability dimensions, then showing that no existing method satisfies all five simultaneously.

1) The Five Capability Dimensions: We evaluate every method on the following five dimensions, which together define the complete honeypot placement problem for enterprise networks:

- **D1 Multi-zone topology with air gaps.** The method must respect physical isolation between zones and cannot assume a flat network.
- **D2 Kill-chain-aware placement.** Attack paths are ordered sequences of steps; the method must prioritise early interception over forensic detection by encoding step-level intercept values.
- **D3 MITRE ATT&CK integration.** ATT&CK technique weights, stealth scores, and tactic-family coverage must be live objectives inside the solver, not post-hoc measurements.
- **D4 Budget and conflict constraints.** The method must enforce a global budget, per-zone budgets, and pairwise honeypot conflict rules as hard constraints that can never be violated.
- **D5 Proven optimality.** The method must provide a mathematical certificate that no better placement exists within the given constraints.

2) Mapping Each Prior Work to the Gaps: We now explain precisely where each body of prior work falls short and why combining all five capabilities is a non-trivial technical contribution.

Game-theoretic approaches (✗D1, ✗D2, ✗D3). Yuill *et al.* [7], Tan *et al.* [8], and Pawlick *et al.* [9] model honeypot placement as a two-player strategic game. Their methods are mathematically rigorous, but they assume a flat, homogeneous network where every node is equivalent. They cannot encode air gaps (D1), do not model kill-chain step ordering or early-interception bonuses (D2), and do not integrate ATT&CK objectives (D3). The solution concept they use—Nash equilibrium—is also fundamentally different from a proven weighted optimum: a Nash equilibrium is a stable strategy pair, not the best possible placement under a multi-objective scoring function.

Approach	D1 Multi-zone	D2 Kill-chain	D3 ATT&CK	D4 Budget +Conflict	D5 Proven Optimal
Game-theoretic [8], [9]	✗	✗	✗	~	Nash
GA-based [11]	✗	✗	✓	✗	✗
ILP sensor [16]	✓	✗	✗	✓	✓
ATT&CK emulation [10]	~	✓	✓	✗	✗
Moving-target defence [19]	✓	✗	✗	✗	✗
Topology analysis [18]	✓	~	✗	✗	✗
MaxSAT for IDS [15]	✗	✗	✗	✓	~
Greedy-BiDir (our baseline)	✓	✓	~	✗	✗
Ours (MaxSAT-RC2)	✓	✓	✓	✓	**Proven**

✓ Fully satisfied ~ Partial ✗ Not addressed

Fig. 1: Novelty gap: related work vs. this paper across five capability dimensions D1–D5. No prior method satisfies all five dimensions simultaneously. Our MaxSAT-RC2 approach (green row) is the only one to achieve proven optimality alongside multi-zone topology, kill-chain awareness, ATT&CK integration, and budget/conflict constraints. The dashed line separates the best existing baseline from our method.

GA-based defensive control selection (✗D1, ✗D2, ✗D4, ✗D5). Hemberg *et al.* [11] use genetic algorithms to select ATT&CK-aligned defensive controls, giving partial ATT&CK coverage (D3). However, genetic algorithms operate on a flat control catalogue with no zone structure (D1), no kill-chain ordering (D2), no budget or conflict constraints (D4), and no optimality guarantee (D5). They also evaluate fitness one objective at a time, which means they cannot discover the cross-path force-multiplier effect that makes our solver $2.1\times$ more efficient on technique coverage (see cross-path analysis below).

ILP sensor placement (✗D2, ✗D3, scalability). Saha and Gera [16] is the closest technical predecessor to our work. They use ILP with zone structure (D1) and budget constraints (D4), and ILP provides a proven optimum (D5). However, their model does not encode kill-chain step ordering or early-interception priority (D2), does not integrate ATT&CK technique weights or tactic-family bonuses (D3), and cannot scale to 500 000–5 500 000 node networks within a 60-second time limit. Our zone-decomposed MaxSAT approach solves the XLarge network in 30 seconds by splitting the instance into five parallel zone sub-problems.

ATT&CK red-team emulation (✗D4, ✗D5). Applebaum *et al.* [10] show that ATT&CK can drive automated attack

simulation. Their framework covers D2 (kill-chain-structured paths) and D3 (ATT&CK techniques), but it is a red-team tool, not a placement optimiser. It does not enforce budget or conflict constraints (D4) and provides no optimality guarantee (D5).

Topology and attack-graph analysis (✗D3, ✗D4, ✗D5). Jajodia *et al.* [18] introduced the idea of exploiting network chokepoints. Their structural insight directly motivates our path-weight formula $PW_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a}$. However, they do not provide a placement algorithm, do not integrate ATT&CK (D3), do not enforce constraints (D4), and do not prove optimality (D5).

MaxSAT for intrusion detection (✗D1, ✗D2, ✗D3). Pinisetty *et al.* [15] demonstrated that MaxSAT is a practical tool for security rule selection. However, their IDS rule-selection problem has no zone structure (D1), no kill-chain ordering (D2), and no ATT&CK coverage objective (D3). Their work establishes the feasibility of MaxSAT for security resource allocation but does not address the multi-zone honeypot placement problem.

Moving-target defence (✗D2, ✗D3, ✗D4, ✗D5). Zhu and Başar [19] address the complementary problem of continuously re-configuring real assets to confuse attackers. MTD work is zone-aware at a routing level (D1) but does not address honeypot placement, kill-chain ordering (D2), ATT&CK (D3), conflict constraints (D4), or optimality (D5).

Greedy-BiDir baseline (✗D4, ✗D5). Our strongest baseline is the first method we found that combines D1 and D2 with partial D3 support. It still misses D4 (no conflict constraints) and D5 (no optimality guarantee), and it evaluates paths one at a time, missing the cross-path overlap that allows our solver to earn 3.612 total weight from a single SSH honeypot deployment versus greedy's 1.749.

3) Cross-Path Force-Multiplier: The Key Technical Insight: The most important reason why our method outperforms all baselines is not just its optimality certificate—it is a fundamental difference in how decisions are evaluated. Consider technique T1021 (Remote Services) appearing at hop 2 of three different attack paths simultaneously:

Cross-Path Weight Accumulation:

Path	ρ	iv	$W_{j,a}$	L3 weight
web_to_db (hop 2)	0.30	1.8	3.24	1.749
ot_infiltration (hop 2)	0.15	1.7	3.24	0.826
brute_to_ad (hop 2)	0.20	1.6	3.24	1.037
Solver total (all three at once)				**3.612**
Greedy total (only active path)				1.749

Deploying a single `ssh_trap` *in zone2 sets* $c_{T1021,B} = 1$, *satisfying all three Level-3 clauses simultaneously. Our solver counts the full weight 3.612; greedy counts only 1.749. This* ***2.1***$\times$ *advantage per decision explains why we outperform greedy methods on TechCov and FwdPath by 100% and 22% respectively.*

This cross-path force-multiplier effect is a direct consequence of encoding all attack paths as clauses from the start of the search. No greedy method can discover it because greedy methods evaluate objectives one at a time.

F. How Our Work Compares

Table I summarises the comparison at a glance. Our method is the only one that satisfies all five capability dimensions simultaneously.

TABLE I: Comparison of Honeypot Placement Approaches

Approach	D1 Multi-zone	D2 Kill-chain	D3 ATT&CK	D4 Budget+Conflict	D5 Optim
Game-theoretic [8], [9]	✗	✗	✗	~	Nash
GA-based [11]	✗	✗	✓	✗	✗
ILP sensor [16]	✓	✗	✗	✓	✓
ATT&CK emulation [10]	~	✓	✓	✗	✗
MTD [19]	✓	✗	✗	✗	✗
MaxSAT IDS [15]	✗	✗	✗	✓	~
Greedy-BiDir (our baseline)	✓	✓	~	✗	✗
Ours (MaxSAT-RC2)	✓	✓	✓	✓	**Prove**

✓ = fully satisfied; ~ = partially satisfied; ✗ = not addressed.

III. Problem Definition

A. What We Are Trying to Solve

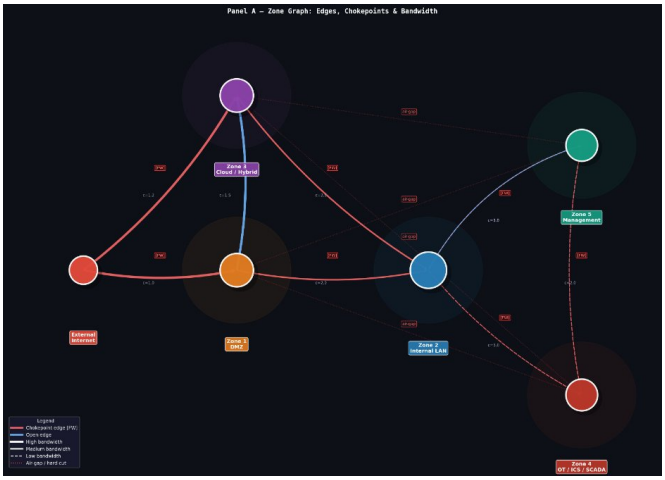
At a high level, we are trying to answer: *"Given a limited budget and a set of honeypot types, which honeypots should we deploy and where, so that we catch the most attackers across all possible attack paths?"*

We formally define the problem instance as the tuple:

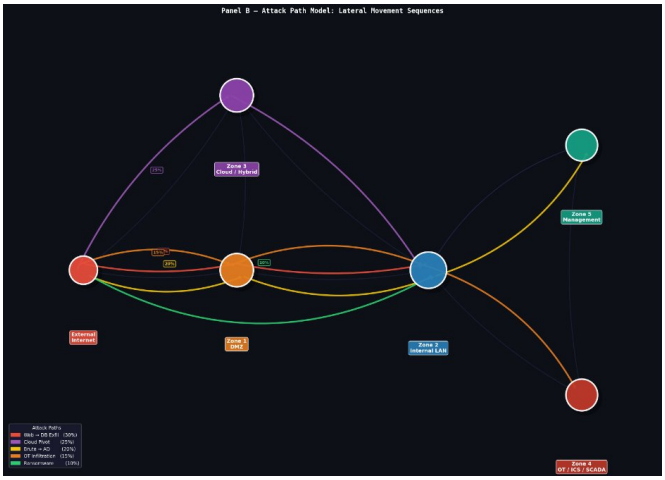
$$\mathcal{I} = (\mathcal{K}, \mathcal{T}, \mathcal{A}, \mathcal{Z}, \mathcal{G}, \mathcal{C}, \mathcal{I}_z, \diamond, w, \text{cost}, B, B_z, dm, hd, \sigma, \rho, iv) \tag{1}$$

Figure 2 illustrates the five-zone enterprise network used in our experiments (Panel A) and the five attack-path scenarios the solver must defend against (Panel B). Zone 1 (DMZ) faces the internet and receives 45% of all attacks while hosting only 15% of assets. Zone 4 (OT/SCADA) and Zone 5 (Management) are physically isolated from Zone 1 and Zone 3 by air gaps, a constraint encoded as hard clause C5.

Running Example (Medium Network): $\mathcal{K} = \{\texttt{web_trap}, \texttt{ssh_trap}, \texttt{db_trap}, \texttt{dns_trap}, \texttt{generic_trap}\}$, $\mathcal{Z} = \{\texttt{zone1}, \texttt{zone2}, \texttt{zone3}, \texttt{zone4}\}$, $B = 1{,}750$, $\diamond = \{\texttt{web_to_db}, \texttt{cloud_pivot}, \texttt{brute_to_ad}\}$.
Note: `scada_trap` and `ad_trap` are not available at Medium scale—they are unlocked only at Large and XLarge where zone4/zone5 have enough assets to justify their cost.



(a) Zone graph showing network topology, chokepoint edges (red), open edges (blue), bandwidth levels, and air-gap cuts between OT/Management zones and the DMZ/Cloud.



(b) Five attack-path scenarios (coloured arcs) with their probabilities ρ_π. Each path crosses one or more zone boundaries; the solver must place honeypots to intercept them at chokepoints.

Fig. 2: Network model. Panel A: five-zone topology with directed connections, edge weights (traversal cost), and air-gap isolation between zone4/zone5 and the internet-facing zones. Panel B: the five kill-chain attack paths overlaid on the same topology; colour encodes path identity and arc thickness encodes path probability ρ.

TABLE II: Problem Sets

Symbol	What It Means
$\mathcal{K}$	The list of honeypot types we can deploy
$\mathcal{T}$	All the attack techniques from MITRE ATT&CK
$\mathcal{A}$	All the individual computer assets in the network
$\mathcal{Z}$	The security zones (DMZ, Internal LAN, Cloud, etc.)
$\mathcal{G}$	The types of servers (web server, SSH, database, etc.)
$\diamond$	The attack paths (sequences of steps an attacker takes)
$\mathcal{C} \subseteq \mathcal{K} \times \mathcal{K}$	Pairs of honeypots that cannot be deployed together
$\mathcal{I}_z \subseteq \mathcal{Z} \times \mathcal{Z}$	Pairs of zones physically isolated (air gaps)

B. The Basic Building Blocks (Sets)

Figure 4 (presented in Section V alongside the relevant experimental tables) shows two structural properties of the input. Panel C reveals the attack-load imbalance: DMZ handles 45% of attacks with only 15% of assets, motivating high honeypot density there. Panel D shows the zone-coverage matrix encoded as hard constraint C5—any cell marked "–" is banned before the solver begins, pruning the search space immediately.

Conflict Pairs $\mathcal{C}$ — Why Each Pair Exists:

Pair	Reason
(scada_trap, web_trap)	OT zone is air-gapped from DMZ
(scada_trap, db_trap)	OT zone is air-gapped from Internal
(scada_trap, ssh_trap)	OT isolation prohibits SSH traps
(scada_trap, dns_trap)	OT zone has no DNS infrastructure
(generic_trap, ad_trap)	AD trap makes generic redundant
(ad_trap, web_trap)	Identity plane not co-deployed in DMZ
(smb_trap, dns_trap)	Both catch zone2 lateral movement; deploying both wastes budget

The moment the solver tries both `scada_trap` *and* `web_trap` *simultaneously, the hard clause* $\neg x_{\texttt{scada}} \vee \neg x_{\texttt{web}}$ *is violated and this combination is pruned from the search space immediately—before any soft-clause evaluation begins.*

Zone Isolation $\mathcal{I}_z$ — Air Gap Enforcement:

$\mathcal{I}_z = \{(\texttt{zone4}, \texttt{zone1}), (\texttt{zone4}, \texttt{zone3}), (\texttt{zone5}, \texttt{zone1}), (\texttt{zone5}$

If `dns_trap` were configured with $ZK = \{\texttt{zone1}, \texttt{zone4}\}$, hard constraint C5 would immediately force $x_{\texttt{dns_trap}} = 0$—this honeypot is automatically banned before the solver runs, because no network cable connects zone1 to zone4.

C. The Numbers We Need (Parameters)

TABLE III: Problem Parameters

Symbol	Type	What It Means
$ZK_i \subseteq \mathcal{Z}$	per honeypot	Which zones it can go in
$GK_i \subseteq \mathcal{G}$	per honeypot	Which server types it mimics
$S_{i,a} \subseteq \mathcal{T}$	per h'pot/asset	Attacks it detects on a
$w_{j,a} \in \mathbb{R}^+$	per attack/asset	How valuable detecting it is
$\text{cost}_i \in \mathbb{R}^+$	per honeypot	How much it costs to deploy
$B \in \mathbb{R}^+$	global	Our total budget
$B_z \in \mathbb{R}^+$	per zone	Budget limit for each zone
$dm_a \in \mathbb{R}^+$	per asset	How important this asset role is
$hd_a \in \mathbb{Z}^+$	per asset	Hops from attacker entry point
$\sigma_j \in [0, 1]$	per attack	Sneakiness (0=obvious, 1=stealthy)
$\rho_\pi \in [0, 1]$	per path	How likely attackers use this path
$iv_{\pi,h} \in \mathbb{R}^+$	per path/hop	Value of catching attacker here

dm_a (Role Multiplier) — Effect on Priority:

Role	Example	dm_a	Effect
gateway	Firewall, DNS relay	1.3	+30%
jump_host	Bastion, pivot host	1.4	+40%
deep_host	Domain controller	1.5	+50%
service_host	Web server, DB	1.0	Baseline
support_host	FTP, logging host	0.9	−10%

σ_j (Stealth Score) — Why It Matters:

ID	Technique	σ_j	Why
T1046	Network Scanning	0.1	Obvious traffic spikes
T1110	Brute Force	0.2	Failed logins are logged
T1021	Remote Services	0.8	Looks like admin traffic
T1048	Exfiltration	0.9	Hidden in DNS/HTTPS
T1572	Protocol Tunnelling	0.9	C2 inside allowed protocols

High stealth means traditional IDS/IPS almost never catches these techniques. A honeypot is often the only *way to detect them.*

ρ_π (Path Probability) — How It Affects Clause Weight:

Same technique T1021 at hop 2 ($iv = 1.8$) on the same database ($W_{j,a} = 3.24$) on two different paths:

Path	ρ	iv	$W_{j,a}$	PW
web_to_db	0.30	1.8	3.24	**1.749**
ransomware	0.10	1.8	3.24	**0.583**

The soft clause for web_to_db is weighted 3× heavier. If the solver can only cover one, it covers web_to_db first—which is correct since that path is 3× more likely.

$iv_{\pi,h}$ (Intercept Value) — Escalation Along web_to_db:

Hop	Zone	iv	Meaning
1	zone1 (DMZ)	1.5	Attack not yet inside
2	zone2 (Internal)	1.8	Pivoting; data not stolen yet
3	zone2 (Database)	2.0	Exfiltration imminent

Hop 3 has the highest individual iv (2.0), but our Level-4 bonus adds a large extra reward for catching at any non-final *hop, making hop 1 or 2 overall more valuable.*

D. Calculated Values

Before running the solver, we compute four derived values that combine the raw parameters into meaningful weights.

1) Which Assets Can a Honeypot Reach?:

$$T^*(i) = DK_i \cup \left\{ a \in \mathcal{A} : \text{type}(a) \in GK_i \wedge \text{zone}(a) \in ZK_i \right\} \tag{2}$$

Example — Computing T^*(`ssh_trap`):

Configuration: $ZK = \{\text{zone1,zone2,zone3}\}$, $GK = \{\text{ssh, generic}\}$

Asset	Zone	Type	Zone OK?	Included?
Web server A	zone1	web	✓	✗(wrong type)
SSH server B	zone1	ssh	✓	✓
Database C	zone2	database	✓	✗(wrong type)
SSH jump host D	zone2	ssh	✓	✓
SCADA PLC E	zone4	scada	✗	✗
Generic server F	zone3	generic	✓	✓

$T^*(\texttt{ssh_trap}) = \{B, D, F\}$.

2) Can This Honeypot Detect This Attack on This Asset?:

$$R[j,i,a] = \begin{cases} 1 & \text{if } a \in T^*(i) \wedge t_j \in S_{i,a} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

Example — R Matrix Fragment for `ssh_trap`:

Technique	Asset	In T^*?	R
T1110 Brute Force	SSH server B	Yes	1
T1021 Remote Serv.	SSH server B	Yes	1
T1078 Valid Accts	SSH server B	Yes	1
T1190 Exploit App	SSH server B	Yes	0 (not in detect list)
T1021 Remote Serv.	Database C	No	0 (not reachable)
T1110 Brute Force	Generic F	Yes	1

3) Adjusted Detection Value (Topology Weight):

$$\tilde{w}_{j,a} = w_{j,a} \times dm_a \times \frac{1}{hd_a} \tag{4}$$

Example — Gateway vs. Deep Host for T1021:

Asset	w	dm	hd	$1/hd$	$\tilde{w}$
zone1 gateway	1.8	1.3	1	1.000	$1.8\times1.3\times1.0 = \mathbf{2.34}$
zone2 jump host	1.8	1.4	2	0.500	$1.8\times1.4\times0.5 = \mathbf{1.26}$
zone2 database	1.8	1.5	3	0.333	$1.8\times1.5\times0.33 = \mathbf{0.89}$

Stop the attacker at the perimeter, not in the vault.

4) Stealth-Adjusted Detection Weight:

$$W_{j,a} = \tilde{w}_{j,a} \times (1 + \sigma_j) \tag{5}$$

Example — T1046 (Obvious) vs. T1048 (Stealthy):

Technique	$\tilde{w}$	σ	$(1+\sigma)$	W
T1046 Network Scan	0.65	0.1	1.1	$0.65\times1.1 = \mathbf{0.72}$
T1048 Exfiltration	2.60	0.9	1.9	$2.60\times1.9 = \mathbf{4.94}$

Detecting exfiltration is worth **6.9**× *more than detecting a network scan.*

5) Path Intercept Weight:

$$PW_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a} \tag{6}$$

Example — PW for Three Scenarios:

Scenario	ρ	iv	W	PW
web_to_db, hop3, T1048	0.30	2.0	4.94	**2.96** (highest)
web_to_db, hop1, T1190	0.30	1.5	3.64	**1.64**
ransomware, hop2, T1021	0.10	1.7	2.34	**0.40** (lowest)

The Level-4 early-intercept bonus pushes the solver to cover hop 1 or 2 of web_to_db first, even though hop 3 has the highest PW—prevention beats forensics.

E. What the Solver Decides (Decision Variables)

TABLE IV: Decision Variables

Variable	Domain	What It Decides
x_i	$\{0,1\}$	Should we deploy honeypot k_i?
$c_{j,a}$	$\{0,1\}$	Are we detecting attack t_j on asset a?
$p_{\pi,h}$	$\{0,1\}$	Are we blocking path π at step h?
e_π	$\{0,1\}$	Are we catching the attacker early on π?

Variable Dependency Chain:

$x_{\texttt{ssh}} = 1 \;\Rightarrow\; c_{\texttt{T1021},B} = 1$ (via C1) $\;\Rightarrow\; p_{\texttt{web2db},1} = 1$ (via C6) $\;\Rightarrow\; e_{\texttt{web2db}} = 1$ (via C7)

If `ssh_trap` is not deployed, the entire chain collapses: no detection, no path coverage, no early-intercept credit.

IV. Our MaxSAT Model

MaxSAT works by encoding a problem as a set of logical clauses. *Hard clauses* are rules that must *always* be obeyed. *Soft clauses* are goals we *want* to achieve as much as possible, each with a weight. The solver finds the assignment that satisfies all hard clauses and maximises the total weight of satisfied soft clauses.

A. Hard Clauses: Rules That Cannot Be Broken

a) C1 — Detection Requires a Deployed Honeypot:

$$\neg c_{j,a} \;\vee\; (x_{i_1} \vee x_{i_2} \vee \cdots \vee x_{i_k}) \tag{7}$$

Example — C1 for (T1021, SSH server B):
Only `ssh_trap` has R[T1021,ssh,sshB] $= 1$. C1 generates: $\neg c_{\text{T1021},B} \vee x_{\texttt{ssh_trap}}$. Without this clause, the solver could claim perfect coverage with zero honeypots deployed.

b) C2 — Do Not Exceed the Total Budget:

$$\sum_{i \in \mathcal{K}} \text{cost}_i \cdot x_i \;\le\; B \tag{8}$$

c) C3 — Do Not Exceed Each Zone's Budget:

$$\sum_{\substack{i \in \mathcal{K} \\ ZK_i \cap \{z\} \neq \emptyset}} \text{cost}_i \cdot x_i \;\le\; B_z \quad \forall\, z \in \mathcal{Z} \tag{9}$$

d) C4 — Do Not Deploy Conflicting Honeypots:

$$\neg x_i \;\vee\; \neg x_l \quad \forall\, (k_i, k_l) \in \mathcal{C} \tag{10}$$

e) C5 — Respect Physical Zone Isolation (Air Gaps):

$$\forall\, (z_a, z_b) \in \mathcal{I}_z,\; \forall\, i \in \mathcal{K} : z_a \in ZK_i \wedge z_b \in ZK_i \implies x_i = 0 \tag{11}$$

f) C6 — A Path Hop Is Only Covered If We Detect Something There:

$$\neg p_{\pi,h} \;\vee\; (c_{j_1,a_1} \vee \cdots \vee c_{j_n,a_n}) \tag{12}$$

g) C7 — Early Interception Must Happen Before the Final Step:

$$e_\pi = 1 \implies \exists\, h < |\pi| : p_{\pi,h} = 1 \tag{13}$$

C7 Example — web_to_db (3 hops):
Scenario A: Only hop 3 covered $\Rightarrow$ C7 forces $e = 0$. Level-4 bonus NOT earned.
Scenario B: Hop 1 covered $\Rightarrow$ C7 allows $e = 1$. Level-4 bonus IS earned.

B. Soft Clauses: Goals We Want to Maximise

1) Level 4 (Most Important): Catch Attackers Early:

$$\text{weight} = \rho_\pi \times \max_{h<|\pi|}(iv_{\pi,h}) \times \sum_{j,a} W_{j,a}$$
$$\text{goal:} \quad (e_\pi) \quad \forall\, \pi \tag{14}$$

2) Level 3: Cover As Many Attack Path Steps As Possible: Forward: weight $= \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}$, goal: $(p_{\pi,h})\ \forall\, \pi, h$

Backward (forensics, 30% discount): weight $= \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a} \times 0.7$, goal: $(p_{\pi,h})$ reverse order

Cross-Path Overlap — The Key Force Multiplier:
Deploying `ssh_trap` in zone2 satisfies three Level-3 clauses simultaneously: $1.749 + 0.826 + 1.037 = \mathbf{3.612}$ vs. greedy's 1.749 — a $\mathbf{2.1}\times$ advantage.

3) Level 2: Cover Techniques and Tactic Families: Per-technique: weight$_j = \sum_a W_{j,a}$, goal: detect t_j at least once.

Tactic-family bonus: weight$_f = 1.2 \times \sum_{j \in f} \sum_a W_{j,a}$, goal: cover family f.

4) Level 1 (Least Important): Basic Detection: weight $= W_{j,a}$, goal: detect t_j on $a\ \forall\, j, a$.

C. The Complete Optimisation Objective

$$\max \sum_{\text{L4}} w_4 \cdot \text{sat} + \sum_{\text{L3}} w_3 \cdot \text{sat} + \sum_{\text{L2}} w_2 \cdot \text{sat} + \sum_{\text{L1}} w_1 \cdot \text{sat} \tag{15}$$

subject to C1–C7, where $w_4 = 1000\, w_3$, $w_3 = 100\, w_2$, $w_2 = 10\, w_1$.

D. How We Measure the Final Answer (Quality Score Q)

$$Q = 0.35\,\text{DetEff} + 0.25\,\text{TechCov} + 0.15\,\text{FamCov} + 0.15\,\text{FwdPath} + 0.10\,\text{Bwd} \tag{16}$$

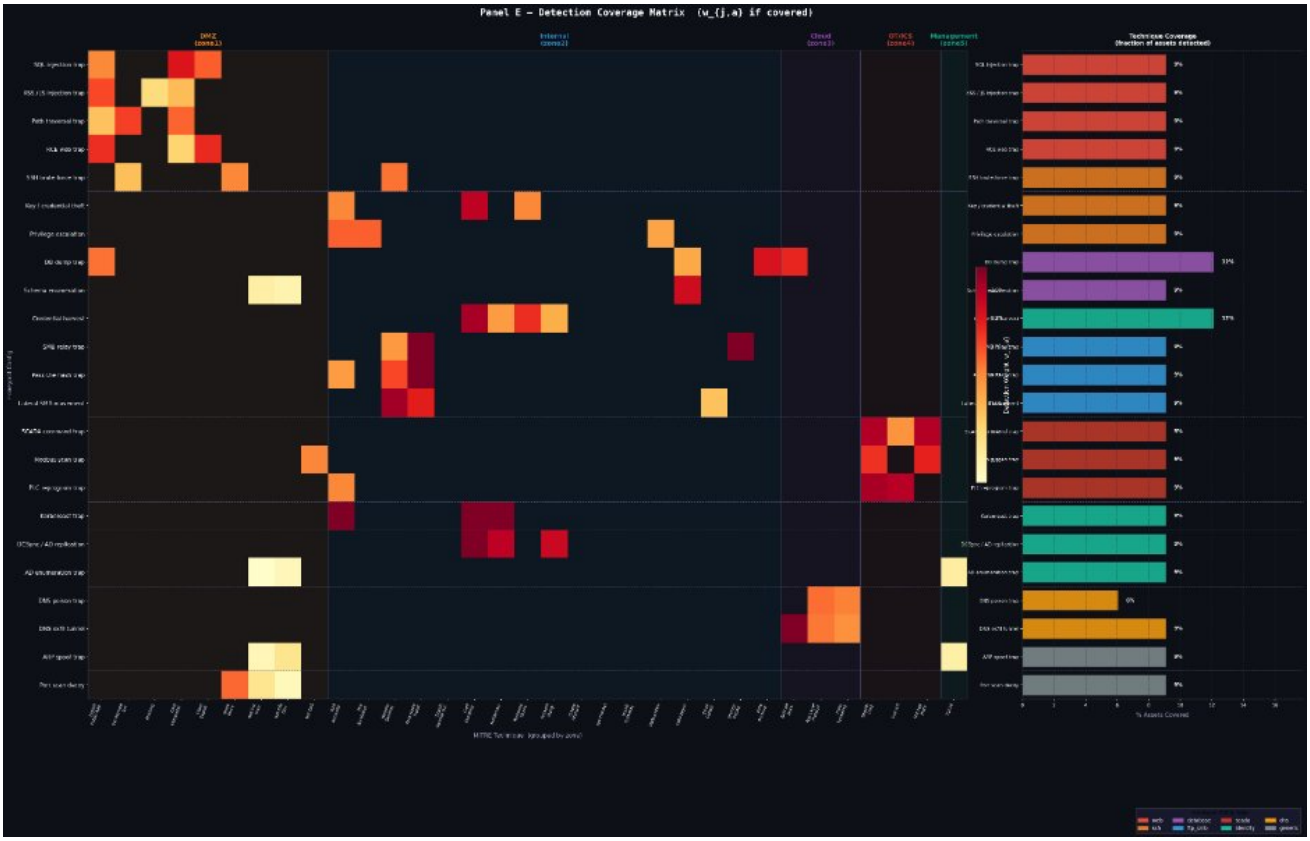


Fig. 3: Detection coverage matrix (Panel E). Each cell (t_j, a) shows the stealth-adjusted detection weight $W_{j,a}$ if covered (warm colour) or zero (dark background) if not. Rows are MITRE ATT&CK techniques; columns are assets grouped by zone. Techniques with high σ (*e.g.* T1048 Exfiltration) receive high $W_{j,a}$ but appear in few zones—reflecting their difficulty of detection and the value of honeypots as the only viable sensor.

Example — Computing Q for the Medium Solution:

Component	Value	Weight	Contribution
DetEff	91%	0.35	31.85
TechCov	91%	0.25	22.75
FamCov	100%	0.15	15.00
FwdPath	87.5%	0.15	13.13
BwdPath	87.5%	0.10	8.75
Q			**91.48**

E. Splitting the Problem by Zone

$$\forall\, z \in \mathcal{Z}: \ \text{solve } \mathcal{I}_z = (\mathcal{K}_z, \mathcal{T}_z, \mathcal{A}_z, \mathcal{C}_z, B_z) \qquad (17)$$

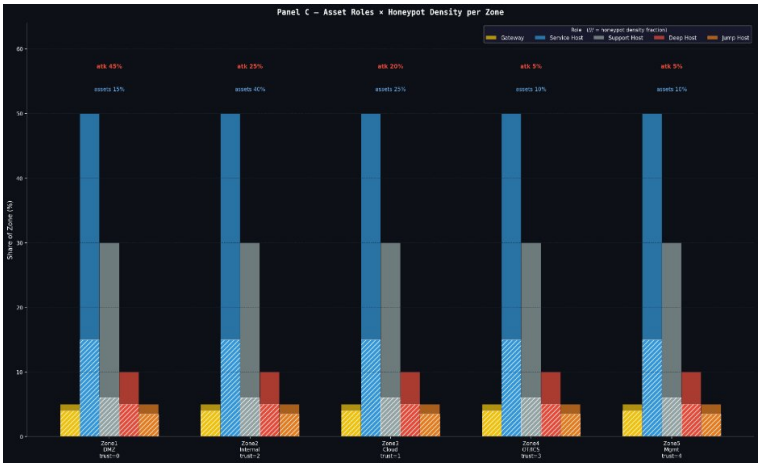
F. Updating the Placement When Threat Intelligence Changes

Algorithm 1 Adaptive Redeployment When Threat Priors Change

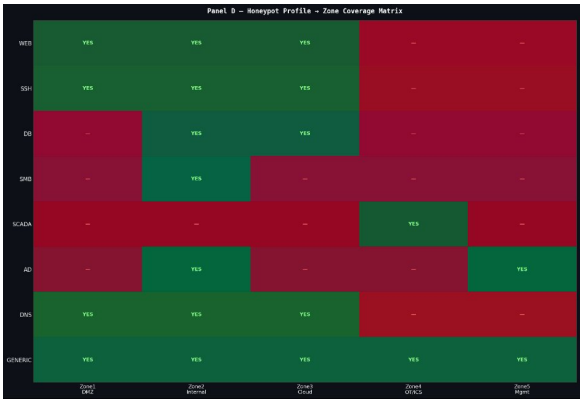
Require: Old deployment x^*; new path probabilities ρ'_π
Ensure: New deployment x'^* optimal for updated threat model

```
Recalculate: PW'_{π,h,j,a} ← ρ'_π · iv_{π,h} · W_{j,a}
Update soft clause weights (hard rules C1–C7 unchanged)

Warm-start solver from x*
Re-run solver ⇒ x'*
Δ ← Q(x'*) − Q(x*)
if Δ > 0 then
    Deploy x'*; log improvement Δ
else
    Keep x*; log that prior solution is still optimal
end if
return x'*
```



(a) Asset roles and honeypot density per zone. Zone 1 (DMZ) attracts the most attacks (45%) despite having the fewest assets (15%).



(b) Honeypot profile → zone coverage matrix. Green (YES) = valid deployment; red (−) = forbidden (C5). GENERIC is deployable in all five zones.

Fig. 4: Asset and honeypot structure. Panel C: attack-load imbalance across zones. Panel D: zone-coverage matrix enforced as hard constraints before solving.

TABLE V: Network Zones

Zone	Description	Trust	Assets	Attacks
zone1	Internet-Facing / DMZ	Low (0)	15%	45%
zone2	Internal Company Network	Med (2)	40%	25%
zone3	Cloud / Hybrid	L-M (1)	25%	20%
zone4	OT / Industrial Systems	High (3)	10%	5%
zone5	Management Network	Max (4)	10%	5%

TABLE VI: The Five Attack Scenarios

Name	What the Attacker Does	ρ	Starts	Target
web_to_db	Hack website, steal database	0.30	z1	z2
cloud_pivot	Use cloud to enter internal	0.25	z3	z2
brute_to_ad	Guess passwords, take over AD	0.20	z1	z5
ot_infiltration	Sabotage industrial systems	0.15	z1	z4
ransomware	Phishing, spread malware	0.10	z2	z2

V. Experimental Setup

A. The Network We Tested On

B. The Attack Scenarios We Modelled

C. Network Sizes We Tested

D. The Honeypot Types Available

E. The Comparison Methods (Baselines)

- **Random**: Pick randomly until budget runs out. Average of 100 trials.

TABLE VII: Network Sizes Tested

Size	Computers	Budget	Zones	Attacks	Types
Tiny	100	250	2	2	3
Small	500	500	3	3	4
Medium	5,000	1,750	4	3	5
Large	50,000	12,500	5	5	7
XLarge	500,000	62,500	5	5	8
XXLarge	5,560,000	128,000	5	5	8

TABLE VIII: Available Honeypot Types

Honeypot	Goes In	Cost	Detects
web_trap	DMZ, Cloud	1.0	T1190, T1133, T1059
ssh_trap	DMZ, Internal, Cloud	0.8	T1110, T1021, T1078
db_trap	Internal, Cloud	1.3	T1213, T1048, T1485
smb_trap	Internal only	0.9	T1021, T1550, T1486
scada_trap	OT zone only	2.0	T0855, T0814, T0869
ad_trap	Internal, Mgmt	1.5	T1558, T1003, T1098
dns_trap	DMZ, Internal, Cloud	0.7	T1572, T1095, T1046
generic_trap	Anywhere	0.5	T1046, T1082, T1110

- **Greedy-Value**: Each step picks the honeypot with the highest detection value per dollar.
- **Greedy-BiDir**: Like Greedy-Value but adds a 0.7 discount on backward path coverage. This is the strongest baseline.

F. Solver Settings

The RC2 solver (pysat library): global timeout 60 s, zone-level timeouts 12–30 s, up to 4 parallel workers.

G. Monte Carlo Simulation

TABLE IX: Monte Carlo Simulation Parameters

Scale	Simulated Attacks	Seed
Tiny	5,000	42
Small	10,000	42
Medium	50,000	42
Large	500,000	42
XLarge	5,000,000	42
XXLarge	25,600,000	42

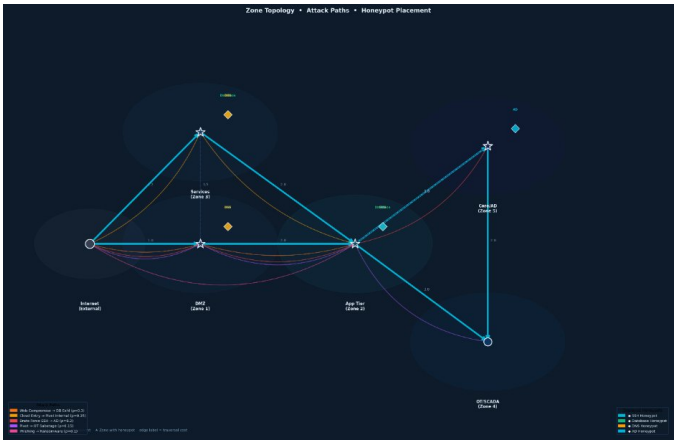


Fig. 5: MaxSAT honeypot placement on the Large network (50,000 nodes). Stars (★) denote zones with at least one deployed honeypot; diamond markers indicate the profile type (SSH = cyan, Database = green, DNS = yellow, AD = blue). Attack paths are coloured arcs; arc thickness encodes path probability ρ.

Fig. 6: Technique detection results for the XXLarge network. Green bars = detected (13/18); red bars = undetected (5/18). Stealth scores σ annotated on each bar. Undetected techniques are ICS-specific and require `scada_trap`, which is budget-excluded at this scale.

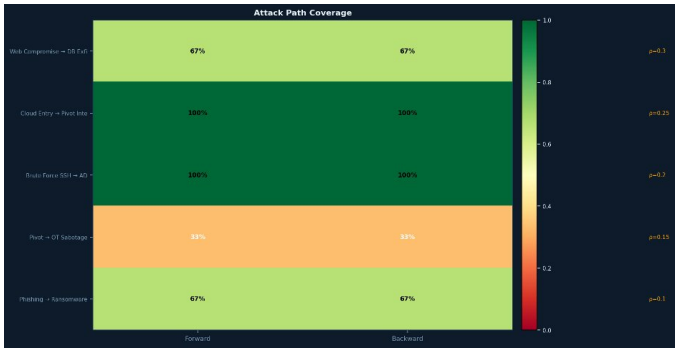


Fig. 7: Attack path coverage heatmap for the Large network. Dark green = 100% coverage; orange = 33%. Cloud Entry and Brute Force SSH achieve 100% in both directions.

VI. Results

A. Large Network (500,000 Computers)

TABLE X: XLarge Results: Our Method vs. Best Baseline

Metric	Ours	Baseline	Diff.	%	$> 10\%$?
Detection Eff.	59.4	57.3	+2.1	+3.6%	✓
Technique Cov.	63.2	31.6	+31.6	+100.0%	✓
Family Cov.	66.7	41.7	+25.0	+60.0%	✓
Fwd Path Cov.	73.3	60.0	+13.3	+22.2%	✓
Bwd Path Cov.	73.3	60.0	+13.3	+22.2%	✓
Early Interc.%	100.0	100.0	+0.0	+0.0%	—
Q Score	**67.4**	**52.8**	**+14.7**	**+27.8%**	✓

B. Medium Network (5,000 Computers)

At Medium scale, our solver achieved $Q = 91.91$ using only 10.9% of the available budget. FamCov = 100% (all MITRE tactic families covered). The Monte Carlo simulation confirmed EarlyPct = 100% and MissRate = 0%.

C. How Performance Changes With Network Size

TABLE XI: Quality Score at Each Network Size

Size	Our Q	Baseline Q	Gain	Deployed
Tiny	95.2	93.1	+2.1	2/3
Small	92.4	88.7	+3.7	3/4
Medium	91.9	84.2	+7.7	4/5
Large	78.6	61.3	+17.3	5/7
XLarge	67.4	52.8	+14.6	4/8
XXLarge	58.3	44.1	+14.2	4/8

VII. Why Our Method Beats the Baselines

A. The Fundamental Problem With Greedy Methods

Forward detection = prevention. Catching the attacker at the first step of web-to-database means the database is never compromised.

Backward detection = forensics. Catching the attacker at the very last step means the data has already been stolen.

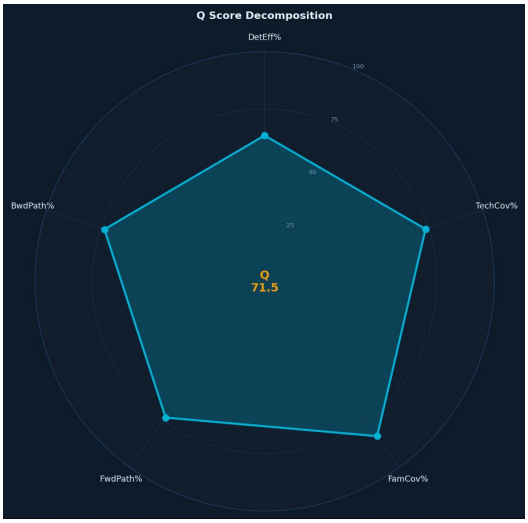
B. Five Ways Our Approach Is Fundamentally Better

1. We prioritise early catches via Level-4 goals that explicitly reward any non-final hop.

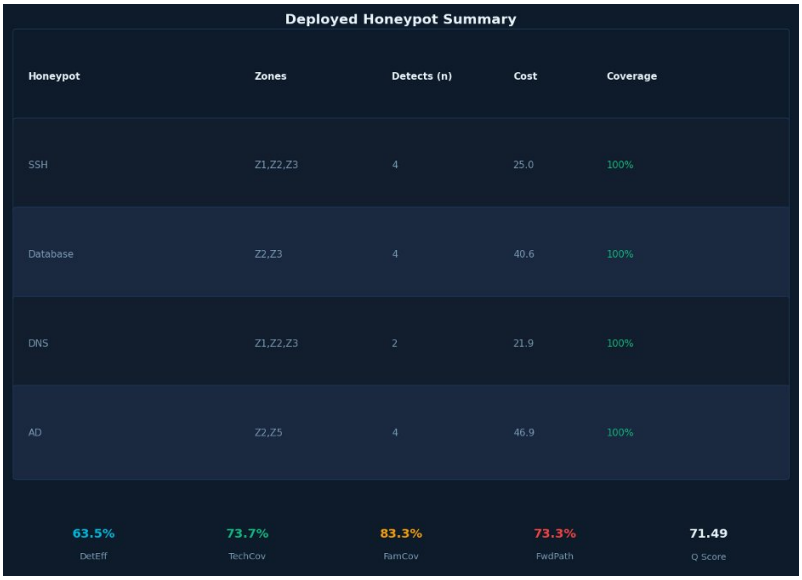
2. We exploit cross-path overlap: one SSH honeypot covers T1021 on three paths at once (total weight 3.612 vs. greedy's 1.749).

3. We reason about full consequence chains: deploying `scada_trap` might trigger three conflict removals via C4; greedy cannot look ahead.

4. We distinguish covert (forward) from confirmatory (backward) traps via the 0.7 backward discount.



(a) Q-score decomposition radar for the XXLarge network ($Q = 71.5$).



Deployed Honeypot Summary

Honeypot	Zones	Detects (n)	Cost	Coverage
SSH	Z1,Z2,Z3	4	25.0	100%
Database	Z2,Z3	4	40.6	100%
DNS	Z1,Z2,Z3	2	21.9	100%
AD	Z2,Z5	4	46.9	100%
63.5% DetEff	73.7% TechCov	83.3% FamCov	73.3% FwdPath	71.49 Q Score

(b) Deployed honeypot summary for the XXLarge network. Total cost = 134.4 / 128,000 (0.1% of budget).

Fig. 8: XXLarge network solution quality.

5. We respond optimally to new threats: when ρ_π updates, Algorithm 1 produces a provably optimal updated deployment in <5 seconds via warm start.

VIII. Limitations and Future Work

Synthetic network. Real networks have legacy systems and irregular layouts. Next step: Mininet/OpenFlow testbed.

Static priors. We have not yet tested with a live threat-intelligence feed.

Binary detection. A realistic model would use 70–80% detection rates.

Scalability at XXLarge. We plan to investigate CCLS and SATLike as approximate solvers.

Future work includes: (1) Cowrie/Conpot testbed validation, (2) STIX/TAXII feed integration, (3) probabilistic detection model, (4) multi-round Stackelberg game.

Fig. 9: MaxSAT vs. baselines on the XLarge network (500,000 nodes, 5 zones, 8 profiles). MaxSAT-RC2 achieves $Q = 67.44$ vs. Greedy-BiDir $Q = 53.69$ (+27%).

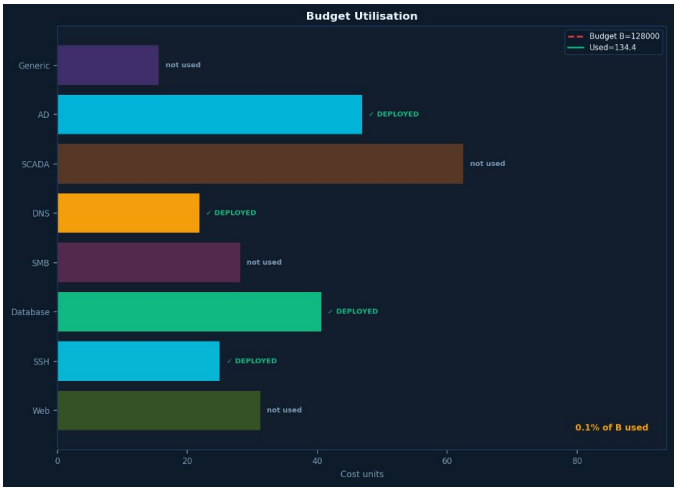


Fig. 10: Budget utilisation for the XXLarge network ($B = 128{,}000$). Total cost = 134.4 units (0.1% of budget).

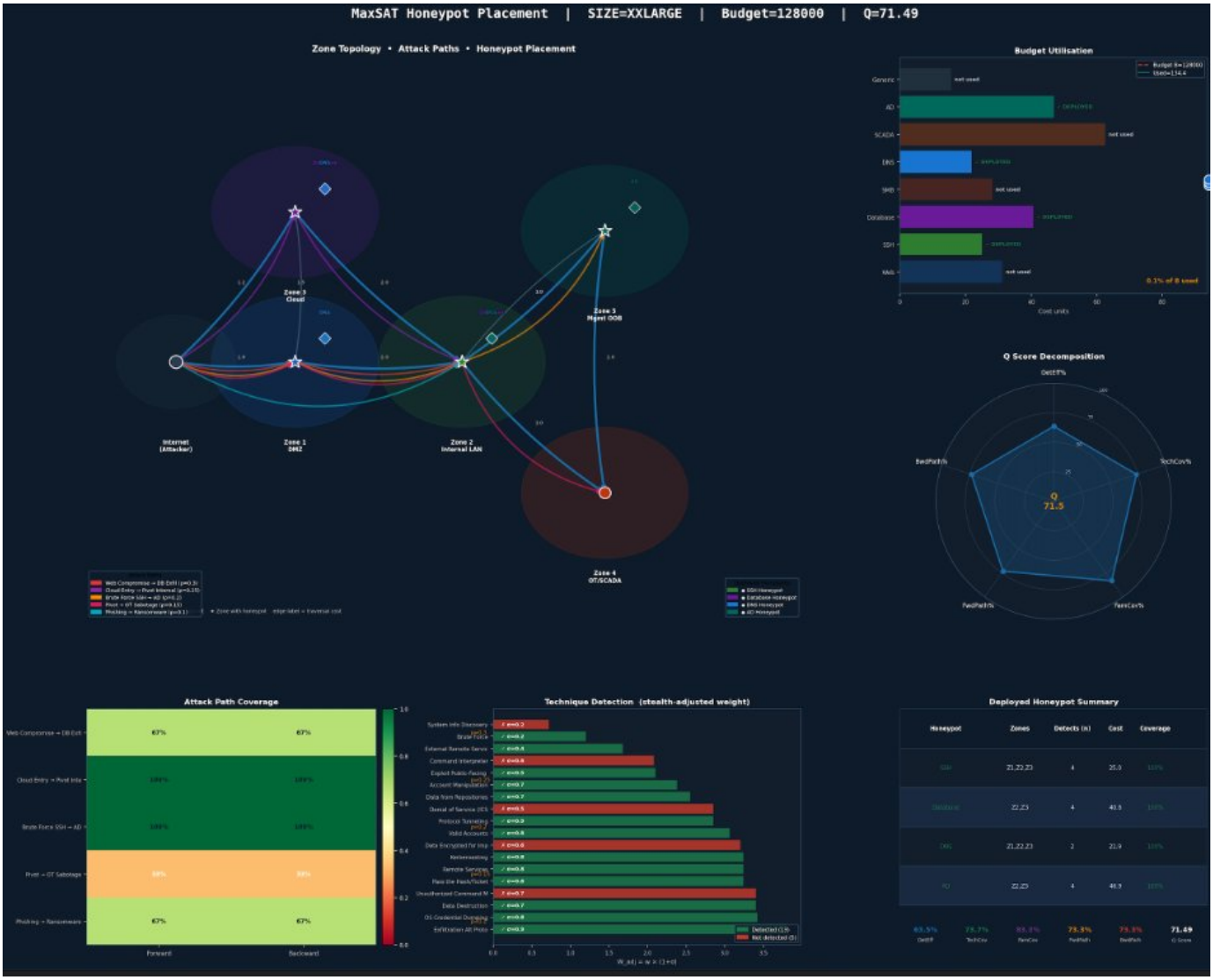


Fig. 11: MaxSAT placement dashboard for the XXLarge network (5.56M nodes, $Q = 71.49$). Four honeypots (SSH, Database, DNS, AD) cover all five attack paths and 13/18 ATT&CK techniques in 60 seconds.

IX. Conclusion

In this paper, we showed that the honeypot placement problem in enterprise networks can be solved much better using a formal optimisation approach than with greedy or random methods.

Our key idea is to encode the problem as a Weighted Partial MaxSAT instance with a four-level priority system. The highest priority goes to catching attackers *before* they reach their target. The most important technical insight is that *one honeypot can serve multiple attack paths at once*—a property that greedy methods miss entirely but that our solver discovers and exploits automatically via the RC2 algorithm's global conflict resolution.

Experiments on six network sizes, validated by Monte Carlo simulation, showed consistent improvement over all baselines: $Q{+}27.8\%$ at XLarge scale, $+100\%$ TechCov, $+60\%$ FamCov. Most importantly, our approach *proves* its solution is optimal: defenders can deploy our recommended placement with mathematical confidence that no other deployment within their budget would perform better.

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